6	**(a)**		$\frac{dT}{dt} = k(T-20)$ [where k is a constant]	**B2** **[2]**	**3.3** **1.1**	May be $-k$ and/or $(20-T)$	B1 if no constant If additional terms seen **B0**
6	**(b)**		k = + 0.07 OR -0.07 oe	**B1**	**1.1**		
			$\int \frac{1}{(T-20)}\, dT = \int k\ dt$	**M1**	**3.1a**	Appropriate separation of variables for *their* DE soi by correct integration or answer Condone missing dt and/or dT	
			$\ln(T-20) = kt\ [+c]$	**M1**	**1.1**	o.e. For correct integration to include a ln term	Or $T-20 = e^{kt\ [+c]}$
			$\ln 70 = c$	**M1**	**1.1**	To find c from their equation using t = 0, T = 90	Or $e^c = 70$
			$\ln(T-20) = -0.07t + \ln 70$				
			$\ln(20) = -0.07t + \ln 70$	**M1**	**3.4**	Clear substitution of $T = 40$ into their equation in T and t coming from integration	$-\frac{2}{7} = -0.07t$
			$t \approx 17.9$ [minutes]	**A1** **[6]**	**3.2a**	17.8966… If there is any doubt about the rigour of an otherwise correct solution withhold the A1.	Any accuracy ≥ 2sf 17.9 nfww gets 6